Python Activity-Based Learning
Program 8: Manufacturing Cost and Profitability Analysis

Problem Statement

Calculate manufacturing cost using material, labour, wastage, and machine costs. Determine the selling revenue and whether the production meets a target profit percentage.

Objectives

• Calculate material wastage cost and total production cost.

• Calculate selling revenue and profit.

• Check whether the required profit target is achieved.

Assumptions

• Material cost is entered for the full batch; wastage is entered as a percentage of material cost.

• Total cost includes material cost, wastage cost, labour, and machine cost.

• Profit percentage is calculated on total production cost.

Algorithm

1. Read product name and production quantity.

2. Read material, labour, and machine costs, wastage percentage, selling price per unit, and target profit percentage.

3. Validate non-negative values and positive quantity.

4. Calculate wastage cost and total production cost.

5. Calculate revenue, profit, and profit percentage.

6. Compare profit percentage with target and display the result.

Flowchart

```
START → Read production details → Validate inputs → Calculate wastage cost → Calculate total cost → Calculate
```

Pseudocode

```
BEGIN
  READ product, quantity, material, labour, machine
  READ wastage_percent, selling_price, target_percent
  IF invalid values THEN DISPLAY error
  wastage_cost = material * wastage_percent / 100
  total_cost = material + wastage_cost + labour + machine
  revenue = quantity * selling_price
  profit = revenue - total_cost
  profit_percent = profit / total_cost * 100
  IF profit_percent >= target_percent THEN DISPLAY target achieved
  ELSE DISPLAY target not achieved
END
```

Python Source Code

```
product = input("Enter product name: ")
quantity = int(input("Enter production quantity: "))
material = float(input("Enter total material cost: "))
labour = float(input("Enter labour cost: "))
machine = float(input("Enter machine cost: "))
wastage_percent = float(input("Enter wastage percentage: "))
selling_price = float(input("Selling price per unit: "))
target_percent = float(input("Target profit percentage: "))

if (quantity <= 0 or material < 0 or labour < 0 or machine < 0
        or wastage_percent < 0 or selling_price < 0
```

```
        or target_percent < 0):
    print("Invalid input.")
else:
    wastage_cost = material * wastage_percent / 100
    total_cost = material + wastage_cost + labour + machine
    revenue = quantity * selling_price
    profit = revenue - total_cost

    if total_cost > 0:
        profit_percent = profit / total_cost * 100
    else:
        profit_percent = 0

    print("\nMANUFACTURING REPORT")
    print("Product:", product)
    print("Wastage cost: Rs.", round(wastage_cost, 2))
    print("Total production cost: Rs.", round(total_cost, 2))
    print("Sales revenue: Rs.", round(revenue, 2))
    print("Profit/Loss: Rs.", round(profit, 2))
    print("Profit percentage:", round(profit_percent, 2), "%")

    if profit_percent >= target_percent:
        print("Target profit achieved.")
    else:
        print("Target profit not achieved.")
```

Test Case

Quantity 100; material ■5,000; labour ■2,000; machine ■1,000; wastage 10%; selling price ■100; target 50%.
Expected: wastage ■500; total cost ■8,500; revenue ■10,000; profit ■1,500; profit percentage about 17.65%; target not achieved.

Sample Output

```
MANUFACTURING REPORT
Wastage cost: Rs. 500.0
Total production cost: Rs. 8500.0
Sales revenue: Rs. 10000.0
Profit/Loss: Rs. 1500.0
Profit percentage: 17.65 %
Target profit not achieved.
```

Conclusion

The program analyses manufacturing costs, wastage, sales revenue, and profit percentage to determine whether the target profit is achieved.